AP CALCULUS AB

Unit 1 • Lesson 1.15 practice • Answer key

Lesson 1.15 • Connecting Limits at Infinity and Horizontal Asymptotes

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

$\lim_{x\to\infty} \dfrac{3x^2+1}{x^2-5}$ is

(A) 3

(B) $-\dfrac{1}{5}$

(C) 0

(D) ∞

Solution

Answer: (A). Equal degrees: $\frac{3}{1} = 3$.

Question 2

$\lim_{x\to\infty} \dfrac{x+1}{x^2}$ is

(A) 1

(B) 0

(C) ∞

(D) $\dfrac{1}{2}$

Solution

Answer: (B). The degree of the denominator is larger.

Question 3

$\lim_{x\to\infty} e^{-x}$ is

(A) 1

(B) ∞

(C) 0

(D) $-\infty$

Solution

Answer: (C). $e^{-x} = \frac{1}{e^x}$ and $e^x \to \infty$.

Question 4

$\lim_{x\to\infty} \dfrac{x^2}{e^x}$ is

(A) ∞

(B) 1

(C) e

(D) 0

Solution

Answer: (D). Exponential growth dominates polynomial growth.

Question 5 (Free response)

Let $f(x) = \dfrac{4x}{\sqrt{9x^2+1}}$.

(a) Find $\lim_{x\to\infty} f(x)$.
(b) Find $\lim_{x\to-\infty} f(x)$. Show why the sign changes.
(c) Write the equations of all horizontal asymptotes of f.
(d) Find $\lim_{x\to\infty} \dfrac{2e^x+5}{e^x-3}$.

Solution

(a) For large $x>0$, $\sqrt{9x^2+1}\approx 3x$, so $f(x)\to \frac{4x}{3x}=\frac{4}{3}$.
(b) For $x<0$, $\sqrt{9x^2}=3|x|=-3x$, so $f(x)\to \frac{4x}{-3x}=-\frac{4}{3}$.
(c) $y=\frac{4}{3}$ and $y=-\frac{4}{3}$.
(d) Divide by e^x: $\dfrac{2+5e^{-x}}{1-3e^{-x}}\to\dfrac{2}{1}=2$.